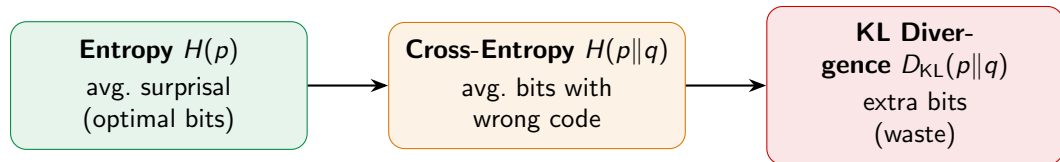


## Information Theory II: ML Connections

KL = MLE · Cross-Entropy Loss · Forward/Reverse KL · Mutual Information

## Recap: The Three Pillars



$$H(p||q) = H(p) + D_{KL}(p||q)$$

**Today:** We connect these concepts to machine learning.  
KL  $\rightarrow$  MLE, cross-entropy  $\rightarrow$  log-loss, forward/reverse KL,  
maximum entropy principle, and mutual information.

## **KL = Maximum Likelihood**

The single most important connection between  
information theory and machine learning.

## Minimizing KL = Maximizing Log-Likelihood

Data from true  $p$ , model family  $q_\theta$ . Find  $\theta$  making  $q_\theta$  closest to  $p$ .

**Step 1:** Write out  $D_{\text{KL}}(p\|q_\theta) = \sum_x p(x)[\log p(x) - \log q_\theta(x)]$ .

*Which of those two terms depends on  $\theta$ ?*

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**Step 3:** Replace  $p$  by empirical  $\hat{p}(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[x_i = x]$ :

$$\arg \max_{\theta} \sum_x \hat{p}(x) \log q_\theta(x) = \arg \max_{\theta} \frac{1}{n} \sum_{i=1}^n \log q_\theta(x_i)$$

$$\arg \min_{\theta} D_{\text{KL}}(p\|q_\theta) = \arg \max_{\theta} \sum_{i=1}^n \log q_\theta(x_i) = \mathbf{MLE!}$$

## Cross-Entropy = MLE: A Concrete Example

Since  $H(p||q_\theta) = H(p) + D_{\text{KL}}(p||q_\theta)$  and  $H(p)$  is constant in  $\theta$ : the same optimum solves all three.

Minimize KL

=

Minimize cross-entropy

=

Maximize log-likelihood

**Numerical example.** 3-class classifier, true class is class 1. Loss is  $-\log_2 \pi_1$ :

| Predicted ( $\pi_1, \pi_2, \pi_3$ ) | Loss (bits) | Interpretation                 |
|-------------------------------------|-------------|--------------------------------|
| (0.7, 0.2, 0.1)                     | 0.51        | confident, correct             |
| (0.4, 0.4, 0.2)                     | 1.32        | uncertain                      |
| (0.1, 0.1, 0.8)                     | 3.32        | wrong, but “not too sure”      |
| (0.05, 0.05, 0.9)                   | 4.32        | <b>confidently wrong!</b>      |
| (1.0, 0.0, 0.0)                     | 0           | confident, correct (saturates) |

Cross-entropy is **harsh on confident wrongness** — that’s the whole point.  
Halving the predicted probability *doubles* the loss.



## Cross-Entropy Loss in Classification

Multi-class classification with  $g$  classes. For observation  $i$ :

- ▶ True label  $y_i \in \{1, \dots, g\}$ , one-hot encoded as  $\mathbf{d}^{(i)} = (0, \dots, 1, \dots, 0)$
- ▶ Model outputs probability vector  $\boldsymbol{\pi}(\mathbf{x}_i \mid \theta) = (\pi_1, \dots, \pi_g)$

The cross-entropy between one-hot  $\mathbf{d}^{(i)}$  and model  $\boldsymbol{\pi}$ :

$$H(\mathbf{d}^{(i)} \parallel \boldsymbol{\pi}) = - \sum_{k=1}^g d_k^{(i)} \log \pi_k(\mathbf{x}_i \mid \theta) = - \log \pi_{y_i}(\mathbf{x}_i \mid \theta)$$

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Summing over all observations:

$$\mathcal{R}(\theta) = - \sum_{i=1}^n \log \pi_{y_i}(\mathbf{x}_i | \theta) \quad (= \text{negative log-likelihood} = \text{log-loss})$$

**Cross-entropy loss IS log-loss IS negative log-likelihood.**

Three names for the same thing. Softmax + cross-entropy = the standard recipe.

## Binary Cross-Entropy = Bernoulli Log-Loss

Binary classification:  $y \in \{0, 1\}$ , model outputs  $\pi(\mathbf{x}) = P(Y=1 \mid \mathbf{x})$ .

$$L(y, \pi) = -y \log \pi(\mathbf{x}) - (1 - y) \log(1 - \pi(\mathbf{x}))$$

This is the cross-entropy between two Bernoulli distributions:  $p = \text{Bern}(y)$  and  $q = \text{Bern}(\pi(\mathbf{x}))$ .

**Connection to MLE / logistic regression:** Logistic regression maximizes likelihood  
 $\Leftrightarrow$  minimizes binary cross-entropy  $\Leftrightarrow$  minimizes KL to the true conditional.  
The cross-entropy loss in logistic regression **is** information-theoretic!

## Maximum Entropy Principle

Given partial knowledge, choose the distribution that is **maximally uncertain** about everything else.

# The Maximum Entropy Principle (Jaynes, 2003)

We know constraints (e.g.,  $\mathbb{E}[g_m(X)] = \alpha_m$  for  $m = 1, \dots, M$ ). Many distributions satisfy them. Which to pick?

**MaxEnt Principle:** of all distributions satisfying the constraints, choose the one with **maximum entropy**.

*Shore & Johnson, 1980:* MaxEnt is the unique inference rule satisfying consistency under coarsening, independence, and continuity.

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**Solution** (Lagrangian):

$$p^*(x) = \frac{1}{Z} \exp\left(\sum_{m=1}^M \lambda_m g_m(x)\right) \quad Z = \sum_x \exp\left(\sum_m \lambda_m g_m(x)\right)$$

The MaxEnt distribution is always an **exponential family** (Gibbs distribution)!

This is no coincidence — exponential families **are** the MaxEnt distributions.

# MaxEnt Recovers Familiar Distributions

Different constraints  $\Rightarrow$  different MaxEnt distributions:

| Constraint(s)                                   | MaxEnt distribution                | Connection             |
|-------------------------------------------------|------------------------------------|------------------------|
| None (fixed support $\{1, \dots, g\}$ )         | Uniform                            | Max entropy = $\log g$ |
| $\mathbb{E}[X] = \mu$ (on $\{1, \dots, g\}$ )   | Exponential/Gibbs                  | Biased die             |
| $\mathbb{E}[X] = \mu, \text{Var}(X) = \sigma^2$ | <b>Gaussian</b> $N(\mu, \sigma^2)$ | Most common in ML!     |
| $\mathbb{E}[X] = 1/\lambda$ (on $[0, \infty)$ ) | <b>Exponential</b> ( $\lambda$ )   | Memoryless             |

**Why the Gaussian is everywhere in ML:** if you only know the mean and variance, the Gaussian is the **least informative** distribution compatible with that knowledge — any other choice smuggles in extra assumptions.

## Proof: Gaussian Maximizes Differential Entropy

**Claim:** Among all densities  $p$  on  $\mathbb{R}$  with mean  $\mu$  and variance  $\sigma^2$ , the Gaussian  $g = N(\mu, \sigma^2)$  maximizes differential entropy.

**Proof** (via Gibbs' inequality applied to differential entropy):

Let  $p$  be any density with the same mean and variance as  $g$ . Consider  $D_{\text{KL}}(p\|g) \geq 0$ :

$$0 \leq D_{\text{KL}}(p\|g) = \int p(x) \log_2 \frac{p(x)}{g(x)} dx = -h(p) - \int p(x) \log_2 g(x) dx.$$

Plug in  $\log_2 g(x) = -\frac{1}{2} \log_2(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2 \ln 2}$  and take the integral. Since  $p$  has the *same* mean and variance as  $g$ , the integral of  $-\log_2 g(x)$  under  $p$  equals its integral under  $g$ :



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$$-\int p(x) \log_2 g(x) dx = -\int g(x) \log_2 g(x) dx = h(g) = \frac{1}{2} \log_2(2\pi e \sigma^2).$$

Substituting back:  $0 \leq -h(p) + h(g) \Rightarrow h(p) \leq h(g)$ . ■

**Same trick works for every MaxEnt distribution** in the table above: pick the candidate distribution, write  $D_{\text{KL}}(p\|\text{candidate}) \geq 0$ , exploit matched moments. The exponential family is essentially the MaxEnt family.

## The CLT, Information-Theoretically

**Classical CLT (recall, Module 20).** Let  $X_1, X_2, \dots$  be i.i.d. with mean 0 and variance  $\sigma^2$ .

The normalized sum

$$Z_n := \frac{X_1 + X_2 + \dots + X_n}{\sqrt{n}} \xrightarrow{d} N(0, \sigma^2).$$

The *distribution* converges to a Gaussian. What does the *entropy* do?

**Use the differential-entropy scaling rule**  $h(aX) = h(X) + \log_2 |a|$ .

With  $S_n = X_1 + \dots + X_n$  (so  $\text{Var}(S_n) = n\sigma^2$ ) and  $a = 1/\sqrt{n}$ :

$$h(Z_n) = h\left(\frac{S_n}{\sqrt{n}}\right) = h(S_n) - \frac{1}{2} \log_2 n.$$

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**Entropic CLT (Barron, 1986):**  $h(Z_n)$  is *monotonically increasing in  $n$*  and converges to the Gaussian's entropy,

$$h(Z_n) \nearrow h(N(0, \sigma^2)) = \frac{1}{2} \log_2(2\pi e \sigma^2).$$

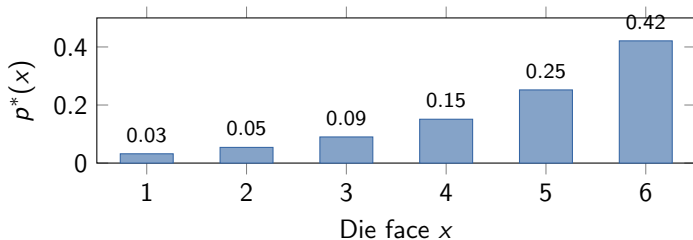
Among distributions with variance  $\sigma^2$ , the Gaussian has **maximum** differential entropy (we proved this). Summation **climbs that hill**: convolution is entropy-increasing, and the Gaussian is its **fixed point** — the entropy attractor.

## MaxEnt Example: The Biased Die

A 6-sided die with  $\mathbb{E}[X] = 4.8$  (heavier than the usual 3.5). What's the MaxEnt distribution?

$$p^*(x) = \frac{e^{\lambda x}}{Z(\lambda)}, \quad Z(\lambda) = \sum_{x=1}^6 e^{\lambda x}, \quad \sum_{x=1}^6 x p^*(x) = 4.8$$

Solving numerically:  $\lambda \approx 0.514$ .



Exponential tilt toward higher faces — the “least opinionated” way to have  $\mathbb{E}[X] = 4.8$ .

**Sign of  $\lambda$  encodes the deviation:**  $\lambda = 0 \rightarrow$  uniform;  $\lambda > 0 \rightarrow$  higher faces;  
 $\lambda < 0 \rightarrow$  lower.

# Mutual Information

How much does knowing  $X$  tell us about  $Y$ ?

A universal measure of dependence.

# Joint and Conditional Entropy

**Joint entropy:** Total uncertainty in  $(X, Y)$  together:

$$H(X, Y) = - \sum_{x,y} p(x, y) \log p(x, y)$$

**Conditional entropy:** Remaining uncertainty in  $Y$  after observing  $X$ :

$$H(Y | X) = - \sum_{x,y} p(x, y) \log p(y | x) = \mathbb{E}_X[H(Y | X=x)]$$

**Two natural questions:**

- ▶ How does  $H(X, Y)$  decompose? (Answer: the *chain rule*, next slide.)
- ▶ How much does  $Y$  *reduce* our uncertainty about  $X$ ? (Answer: *mutual information*, in 2 slides.)

Joint and conditional entropy will give us the two key identities of the rest of this lecture.

## Chain Rule: The Workhorse Identity

The chain rule **factors** joint uncertainty into stages. Both directions are valid:

$$H(X, Y) = H(X) + H(Y | X) = H(Y) + H(X | Y).$$

**Symmetry consequence:**  $H(X) - H(X|Y) = H(Y) - H(Y|X)$ . (This will be *mutual information* in 2 slides.)

**Worked example:** two coin flips  $X, Y \in \{0, 1\}$ , both fair, with  $Y = X \text{ XOR } Z$  where  $Z \sim \text{Bern}(0.1)$  independent of  $X$ .

- $H(X) = 1$  (fair coin).
- $H(Y|X) = H(Z) = h(0.1) \approx 0.469$  (binary entropy of 0.1).
- $H(X, Y) = H(X) + H(Y|X) \approx 1.469$  bits.

*Sanity check the other direction:*  $Y$  marginally is also fair ( $\Rightarrow H(Y) = 1$ ), and by symmetry  $H(X|Y) \approx 0.469$ . Same total. ✓

**$n$ -variable generalization:**  $H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i | X_1, \dots, X_{i-1})$ .  
This is exactly how autoregressive language models factor the joint probability of a sequence.

## Mutual Information: Three Equivalent Definitions

**Question:** How much does knowing  $X$  **reduce** our uncertainty about  $Y$ ?

**Form 1 — uncertainty reduction:**

$$I(X; Y) = H(X) - H(X | Y).$$

*Knowing  $Y$  shrinks our uncertainty about  $X$  by this much.*



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**Form 2 — by symmetry of the chain rule:**

$$I(X; Y) = H(Y) - H(Y | X) = H(X) + H(Y) - H(X, Y).$$

*MI is symmetric in  $X$  and  $Y$  — unlike KL.*

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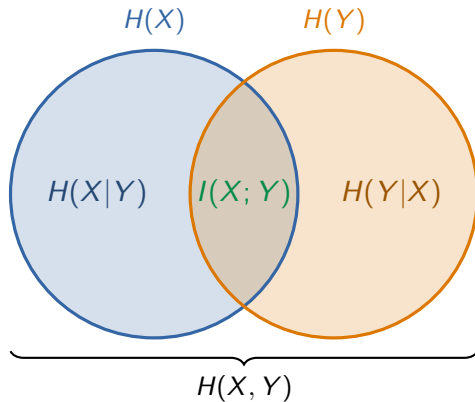
**Form 3 — KL from independence:**

$$I(X; Y) = D_{\text{KL}}(p(x, y) \parallel p(x)p(y)) = \sum_{x, y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}.$$

*MI = how far the joint is from the product of marginals.*

$I(X; Y) = 0$  iff  $X \perp\!\!\!\perp Y$ . The larger  $I$ , the more  $X$  and  $Y$  “know” about each other.

## The Information Diagram



$I(X; Y) = H(X) + H(Y) - H(X, Y)$  — the overlap = shared information.

## Properties of Mutual Information

**1. Non-negative:**  $I(X; Y) \geq 0$  (since  $D_{\text{KL}} \geq 0$ )

**2. Zero iff independent:**  $I(X; Y) = 0 \Leftrightarrow X \perp\!\!\!\perp Y$

**3. Symmetric:**  $I(X; Y) = I(Y; X)$  (unlike KL!)

**4. Self-information:**  $I(X; X) = H(X)$

**5. Bounded:**  $I(X; Y) \leq \min\{H(X), H(Y)\}$  (for discrete RVs)

**6. Invariant:** under invertible smooth transformations of  $X$  or  $Y$

**Conditioning reduces entropy (“information can’t hurt”):**

$H(X | Y) \leq H(X)$ , with equality iff  $X \perp\!\!\!\perp Y$ .

Proof:  $0 \leq I(X; Y) = H(X) - H(X | Y)$ .

## Worked Example: Computing MI

**Joint distribution**  $p(x, y)$ :

| $X \backslash Y$ | $y_1$          | $y_2$          | $y_3$          | $y_4$          |
|------------------|----------------|----------------|----------------|----------------|
| $x_1$            | $\frac{1}{8}$  | $\frac{1}{16}$ | $\frac{1}{32}$ | $\frac{1}{32}$ |
| $x_2$            | $\frac{1}{16}$ | $\frac{1}{8}$  | $\frac{1}{32}$ | $\frac{1}{32}$ |
| $x_3$            | $\frac{1}{16}$ | $\frac{1}{16}$ | $\frac{1}{16}$ | $\frac{1}{16}$ |
| $x_4$            | $\frac{1}{4}$  | 0              | 0              | 0              |

**Marginals:**  $p_X = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ ,  $p_Y = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$ .

**Group the joint by value** (read off the table):

$$\blacktriangleright 1 \times \frac{1}{4}, \quad 2 \times \frac{1}{8}, \quad 6 \times \frac{1}{16}, \quad 4 \times \frac{1}{32}$$

$$H(X) = \log_2 4 = 2 \text{ (uniform).}$$

$$H(Y) = \frac{1}{2}(1) + \frac{1}{4}(2) + 2 \cdot \frac{1}{8}(3) = \frac{7}{4}.$$

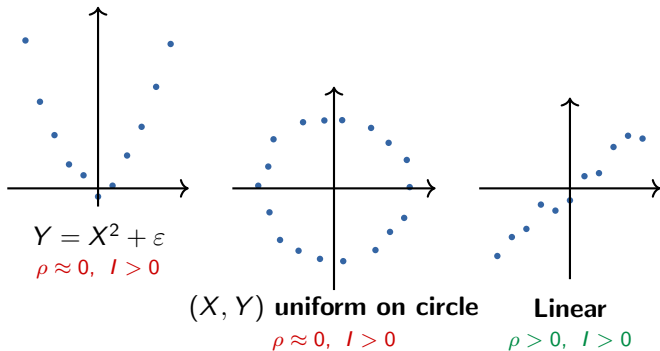
$$\begin{aligned} H(X, Y) &= \frac{1}{4}(2) + 2 \cdot \frac{1}{8}(3) + 6 \cdot \frac{1}{16}(4) + 4 \cdot \frac{1}{32}(5) \\ &= \frac{4}{8} + \frac{6}{8} + \frac{12}{8} + \frac{5}{8} = \frac{27}{8}. \end{aligned}$$

**Mutual information:**

$$I(X; Y) = 2 + \frac{7}{4} - \frac{27}{8} = \boxed{\frac{3}{8} \text{ bits}}$$

## MI vs Correlation: Why MI Is More General

**Pearson correlation** only measures **linear** dependence. MI captures **any** dependence.



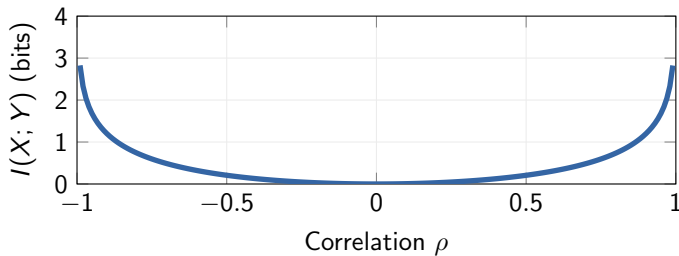
**Correlation = 0 does NOT mean independence.** MI detects **any** dependence — linear, nonlinear, or otherwise.

## MI for Correlated Gaussians

For any  $(X, Y) \sim N\left(\boldsymbol{\mu}, \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}\right)$ :

$$I(X; Y) = -\frac{1}{2} \log_2(1 - \rho^2).$$

Why:  $h(X) + h(Y) - h(X, Y) = \frac{1}{2} \log(2\pi e \sigma_X^2 \cdot 2\pi e \sigma_Y^2) - \frac{1}{2} \log((2\pi e)^2 \sigma_X^2 \sigma_Y^2 (1 - \rho^2))$ . All but  $\rho$  cancels.

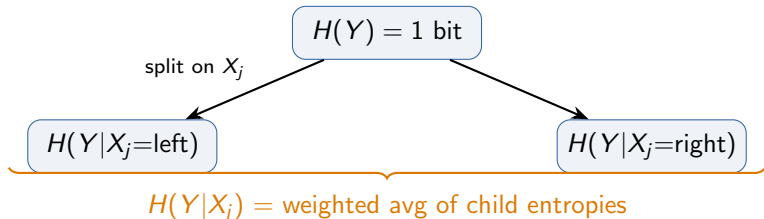


$\rho = 0 \Rightarrow I = 0$  (Gaussian: uncorrelated = independent).  $|\rho| \rightarrow 1 \Rightarrow I \rightarrow \infty$ .

## MI in ML: Information Gain in Decision Trees

At each node of a decision tree, we pick the feature that **maximizes information gain**:

$$\text{IG}(Y; X_j) = H(Y) - H(Y | X_j) = I(Y; X_j)$$



$$\text{IG} = H(Y) - H(Y|X_j) = I(Y; X_j)$$

**Split on the feature with highest MI with the target.**

Exactly the criterion in ID3 and C4.5. CART uses Gini by default but can use entropy. Feature selection: rank features by  $I(X_j; Y)$ , pick top ones.



# The Information Theory Toolbox for ML

| Concept                       | Formula                            | ML Role                         |
|-------------------------------|------------------------------------|---------------------------------|
| Entropy $H(p)$                | $-\sum p \log p$                   | Baseline risk, impurity         |
| Cross-entropy $H(p\ q)$       | $-\sum p \log q$                   | <b>Loss function</b> (log-loss) |
| KL divergence $D_{\text{KL}}$ | $\sum p \log \frac{p}{q}$          | MLE, variational inference      |
| Mutual info $I(X; Y)$         | $D_{\text{KL}}(p_{xy} \  p_x p_y)$ | Feature selection, info gain    |
| Forward KL                    | $D_{\text{KL}}(p \  q_\theta)$     | Supervised learning             |
| Reverse KL                    | $D_{\text{KL}}(q_\phi \  p)$       | Variational inference           |
| MaxEnt                        | $\arg \max H$ s.t. constraints     | Gaussian, exp family            |

**Key takeaway:** Information theory is not just “theory” — it directly gives us the loss functions, optimization objectives, and model selection criteria we use every day in ML.

# Questions?